ITA 0608 - MACHINE LEARNING

1. Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

Aim

To implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

Algorithm

1. Initialize the hypothesis as the most specific hypothesis (NULL).
2. Read the training examples.
3. Consider only the positive training examples.
4. For the first positive example, set the hypothesis equal to it.
5. For each subsequent positive example:
    - Compare each attribute with the hypothesis.
    - If both are the same, keep the attribute.
    - Otherwise replace it with ?.
6. Ignore all negative examples.
7. Display the final hypothesis.

Code

```python
import pandas as pd
data = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
]
columns = ['Sky', 'AirTemp', 'Humidity', 'Wind', 'Water', 'Forecast', 'EnjoySport']
df = pd.DataFrame(data, columns=columns)
print("Training Data:")
print(df)
hypothesis = None

for i in range(len(df)):
    if df.iloc[i, -1] == "Yes":
        if hypothesis is None:
            hypothesis = list(df.iloc[i, :-1])
        else:
            for j in range(len(hypothesis)):
                if hypothesis[j] != df.iloc[i, j]:
                    hypothesis[j] = '?'

print("\nFinal Hypothesis:")
print(hypothesis)
```

Output

```
Training Data:
     Sky AirTemp Humidity    Wind Water Forecast EnjoySport
0  Sunny    Warm   Normal  Strong  Warm     Same        Yes
1  Sunny    Warm     High  Strong  Warm     Same        Yes
2  Rainy    Cold     High  Strong  Warm   Change         No
3  Sunny    Warm     High  Strong  Cool   Change        Yes

Final Hypothesis:
['Sunny', 'Warm', '?', 'Strong', '?', '?']
```

+ Code + Text

Result

The FIND-S algorithm was successfully implemented. The most specific hypothesis obtained from the given training data is:

['Sunny', 'Warm', '?', 'Strong', '?', '?']

This hypothesis represents the most specific concept that is consistent with all the positive training examples while ignoring the negative examples.